

A. Group Information

B. **Group Number** _____, **19** _____

Date: 29-01-2026

S.NO	Name of Student	Roll Number	Signature
01	Shazil Ahmed Shaikh	22CS026	
02	Syed Yashir Raza Zaidi	22CS020	
03	Syed Abdul Ghani Shah	22CS110	

Note: The name of Group Leader should be written at S.No. 01

B. Project Information

1. Major Domain:

Artificial Intelligence (AI), Extended Reality (XR), Human-Computer Interaction (HCI)

2. Proposed Title: AI-Powered Extended Reality (XR) Assistive Guidance System for Real-Time Task Support

3. Project Type:

Software (AI + XR) with optional hardware support (AR glasses / mobile devices)

C. Sustainable Development Goals (SDGs) Covered

This project aligns with the following United Nations Sustainable Development Goals (SDGs):

- **SDG 3 – Good Health and Well-being:** Assistive XR guidance for fitness training, rehabilitation, and elderly support.
- **SDG 4 – Quality Education:** Interactive AR-based learning and step-by-step guidance for digital literacy and skill development.
- **SDG 9 – Industry, Innovation, and Infrastructure:** Promotes innovation through AI-driven XR systems.
- **SDG 11 – Sustainable Cities and Communities:** Disaster management and emergency response guidance through XR.
- **SDG 10 – Reduced Inequalities:** Improves accessibility for elderly and non-technical users.

D. Characteristics of CEP (Complex Engineering Problem)

The project addresses multiple CEP characteristics:

- Multidisciplinary integration of AI, Computer Vision, XR, and HCI
- Requirement of real-time perception and decision-making
- Interaction with uncertain and dynamic environments
- Consideration of human safety, usability, and accessibility
- Design under practical constraints (latency, hardware limits)

E. Background

With the rapid advancement of Artificial Intelligence and Extended Reality (XR), there is a growing opportunity to bridge the gap between complex technology and human usability. Many users—especially elderly individuals, beginners, or people in high-stress environments—struggle to understand modern interfaces or perform physical tasks correctly. Existing instructional systems rely heavily on text or videos, which lack real-time contextual awareness.

AI-powered XR systems can observe the real world, understand the environment, and overlay meaningful guidance directly onto the user's field of view. By combining scene understanding, reasoning, and immersive visualization, such systems can provide intuitive, step-by-step assistance for a wide range of real-world tasks.

F. Problem Statement

Users often face difficulty performing tasks that require procedural knowledge, physical precision, or interaction with complex interfaces. Traditional instructions are static and fail to adapt to real-world context. There is a lack of intelligent, real-time assistive systems that can perceive the environment, understand user intent, and provide actionable guidance directly within the user's view.

G. Aim & Objectives

Aim:

To develop an AI-powered XR system that provides real-time, context-aware visual guidance for task execution across multiple domains.

Objectives:

- To design a scene understanding pipeline using computer vision and AI
- To integrate AI reasoning for task interpretation and guidance generation
- To overlay AR-based visual cues such as arrows, highlights, and 3D models
- To support voice-based interaction and accessibility features

- To evaluate usability, accuracy, and task completion efficiency
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H. Scope

In Scope:

- Real-time detection of objects, screens, or tools
- AI-driven decision-making for guidance generation
- AR overlays including arrows, highlights, and 3D demonstrations
- Cloud-based AI processing with XR visualization
- Use-case-driven prototypes

Out of Scope:

- Full offline/on-device AI processing
 - High-end commercial AR hardware development
 - Medical diagnosis or safety-critical certifications
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I. Proposed Use Cases

1. AI XR Gym Coach

- Detects gym equipment (e.g., treadmill, bench press)
- Displays a 3D human model performing the correct exercise
- Provides posture correction and repetition guidance in AR

2. AI XR Cooking Assistant

- Recognizes kitchen tools and ingredients
- Displays step-by-step recipe instructions in AR
- Shows 3D animations for cutting, mixing, and cooking techniques

3. AI XR Digital Assistant for Elderly Users

- Understands computer screens and UI elements (e.g., MS Word)
- Guides users using AR arrows and voice prompts
- Assists in completing digital tasks easily

4. AI XR Disaster Management Assistant

- Provides real-time guidance during emergencies
 - Shows safe evacuation paths and instructions in AR
 - Assists rescue workers with spatial awareness and task execution
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J. High-Level System Architecture

1. Camera captures real-world scene
 2. Computer Vision detects objects/UI elements
 3. AI model understands context and user intent
 4. LLM-based reasoning generates guidance steps
 5. XR engine renders visual overlays
 6. Voice feedback assists the user
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K. Tools & Technologies (Summary)

- **XR Engine:** Unity + AR Foundation
 - **AI & CV:** OpenCV, OCR, YOLO (optional)
 - **AI Reasoning:** Cloud-based LLMs
 - **Programming:** C#, Python
 - **Hardware:** AR-enabled smartphone / AR glasses
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Note

Any change in the name of supervisor, project title, or group members by the students should be notified to TRC.